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A $I = nAve$

$$v = \frac{I}{nAe} = \frac{I}{ne\pi\frac{d^2}{4}} = \frac{4I}{ne\pi d^2}$$

Since I , n and e are constants, as x increases, d decreases and v increases.
The graph is non-linear as v is not linearly related to d .

diode

resistor

filament lamp

